

Purpose

Build a reusable software tone generator that converts an active button, switch, or keypad into a musical note

Standards & References

- RCET 3375 Lab standard
- RCET PIC-AS style guide
- RCET Flowchart guide
- PIC16F883 Data sheet
- PIC Micro Mid-Range MCU Family Manual
- Previous Lab documentation for delays & Keypad Scanning

Equipment & materials

- MPLAB XIDE & PIC-AS toolchain
- PICkit programmer/debugger
- PIC16F883 circuit
- O-Scope & frequency counter
- 8 position dip switch & 4x4 matrix keypad
- Speaker & class D driver circuit
- Lab book & components as required

Class D driver

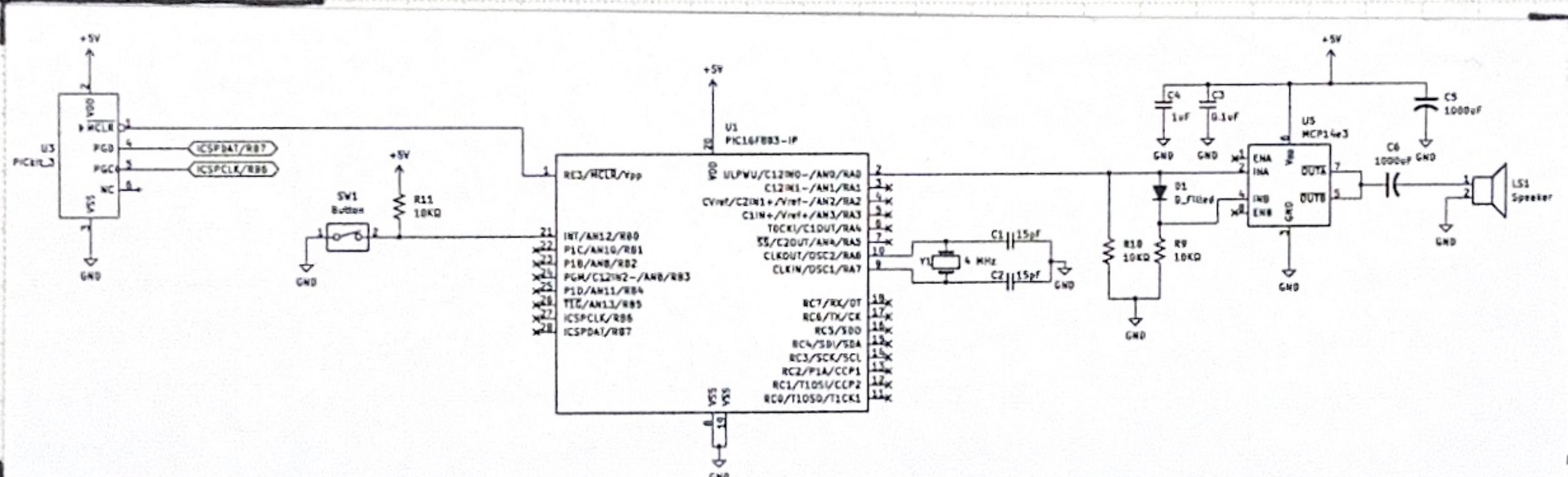


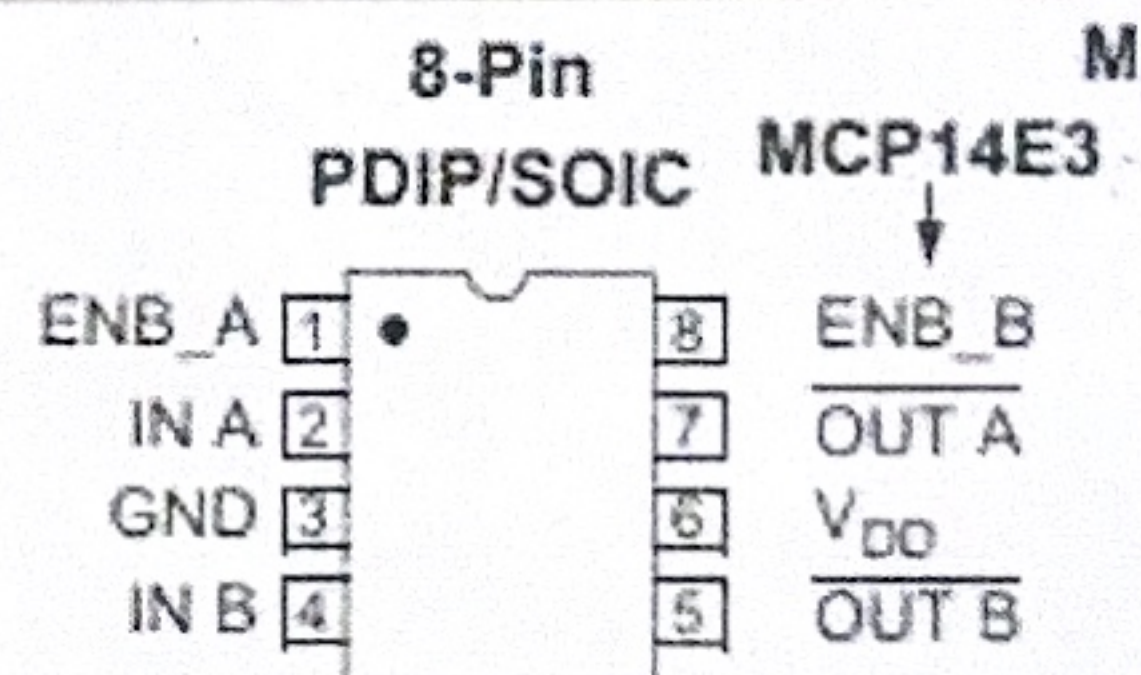
Figure 1-1
PIC16F883
with Driver
Circuit

*The MCP14e3 Class D driver can operate at up to 18V & an input current of 50mA. The PIC16F883 at max could output 25mA. In our situation there is 5V being sunk through two 10K's in parallel so a max of 1mA will be active at worst. All loading is time

Absolute Maximum Ratings†

Supply Voltage	+20V
Input Voltage	(V _{DD} + 0.3V) to (GND - 5V)
Enable Voltage	(V _{DD} + 0.3V) to (GND - 5V)
Input Current (V _{IN} > V _{DD})	50 mA
Package Power Dissipation (T _A = 50°C)	Note 3
8L-DFN-S	1.10W
8L-PDIP	665 mW
8L-SOIC	

MCP14e3 Absolute max



MCP14e3 Pinout

Part 1 - Generate & Amplify a Single Tone

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Generate a nominal 1KHz, with a 50% duty cycle Square wave & reproduce it only while a button is actively pressed. Drive the speaker through the provided D class circuit

Before Lab

Start with Software timing knowledge & reusable delay work. prepare the circuit & program so that:

- The tone output is logic low when the button isn't pressed
- The button state is read continuously
- the used delay is implement as reusable logic rather than embedded repeatedly in the main loop

Prepare or reference:

- Schematic & class D interface documentation
- applicable PIC SFR documentation (pg 308)
- Loading calcs
- Top Level flow chart
- timing calculations
- Source code

Main:

```
BTFSS PORTB,0
goto NoButton
```

} 2 when set

DelayLoop:

```
MOVLW 0x01 ;1 TCY code not part of delay
XORWF 0x05,1 ;1 TCY code not part of delay
```

} 2 TCY

```
MOVLW 0xA3 ;1 TCY delay overhead
MOVWF count ;1 TCY delay overhead
```

} 2 TCY

CALL Delay

} 2 TCY

goto Main

} 2 TCY

Delay:

```
DECFSZ count,1 ;1(2) 1 TCY if not zero, 2 TCY if zero
goto Delay ;2 TCY
```

} 3(I-1)+2

RETURN

} 2 TCY

NoButton:

```
MOVLW 0x00
MOVWF 0x05
goto Main
```

Code 1-1 1KHz tone delay loop

~~3I+9~~

$$3I+11 \rightarrow 3(163)+11 = 500\mu s = PW$$

Best I=163

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In the Lab

1. Build & program the circuit
2. Verify the tone doesn't play when the button isn't pressed
3. Press & hold the button & verify the tone doesn't stop
4. Verify PW, PS, Period, freq, & duty cycle with the O-scope & freq counter
5. document troubleshooting

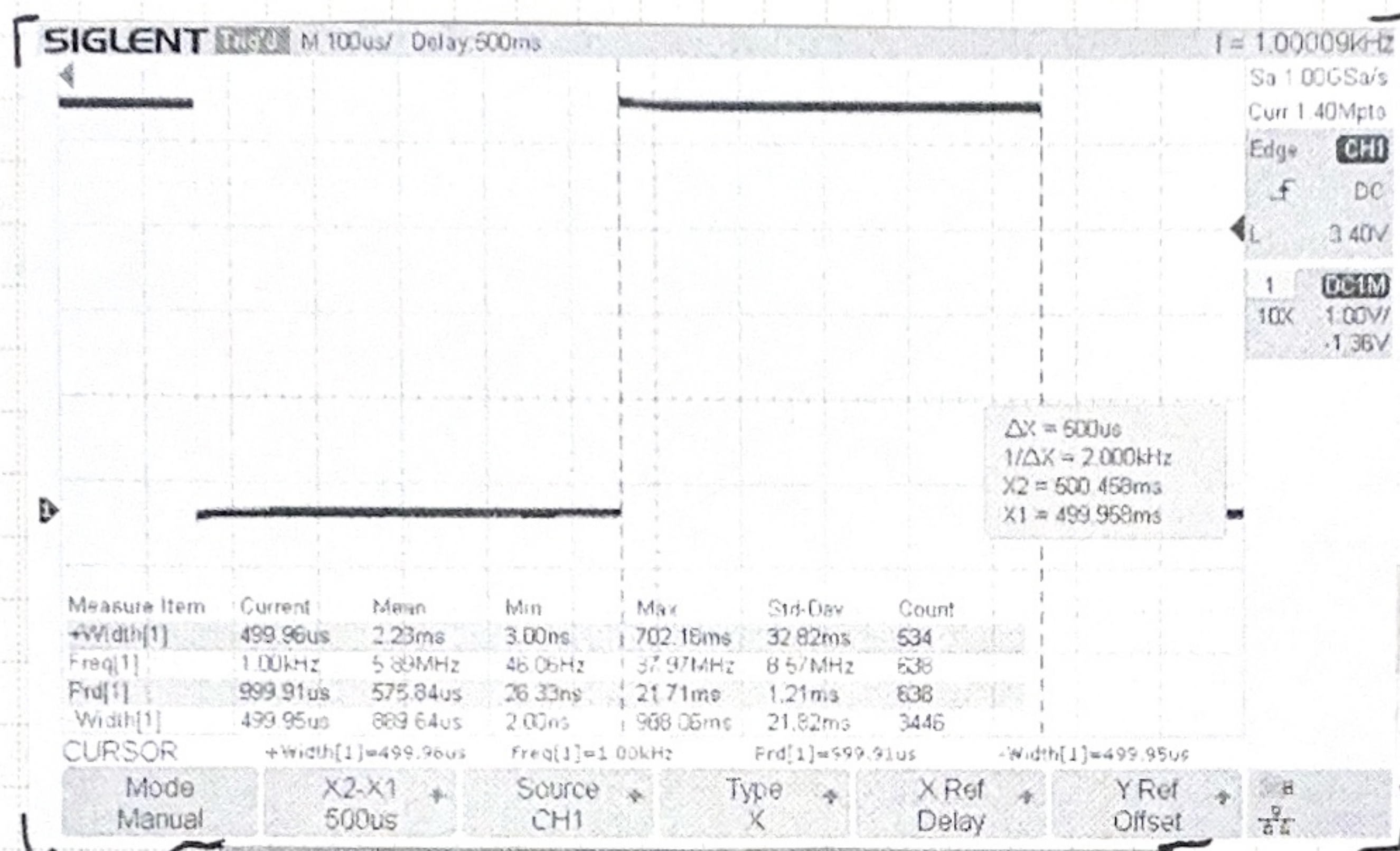
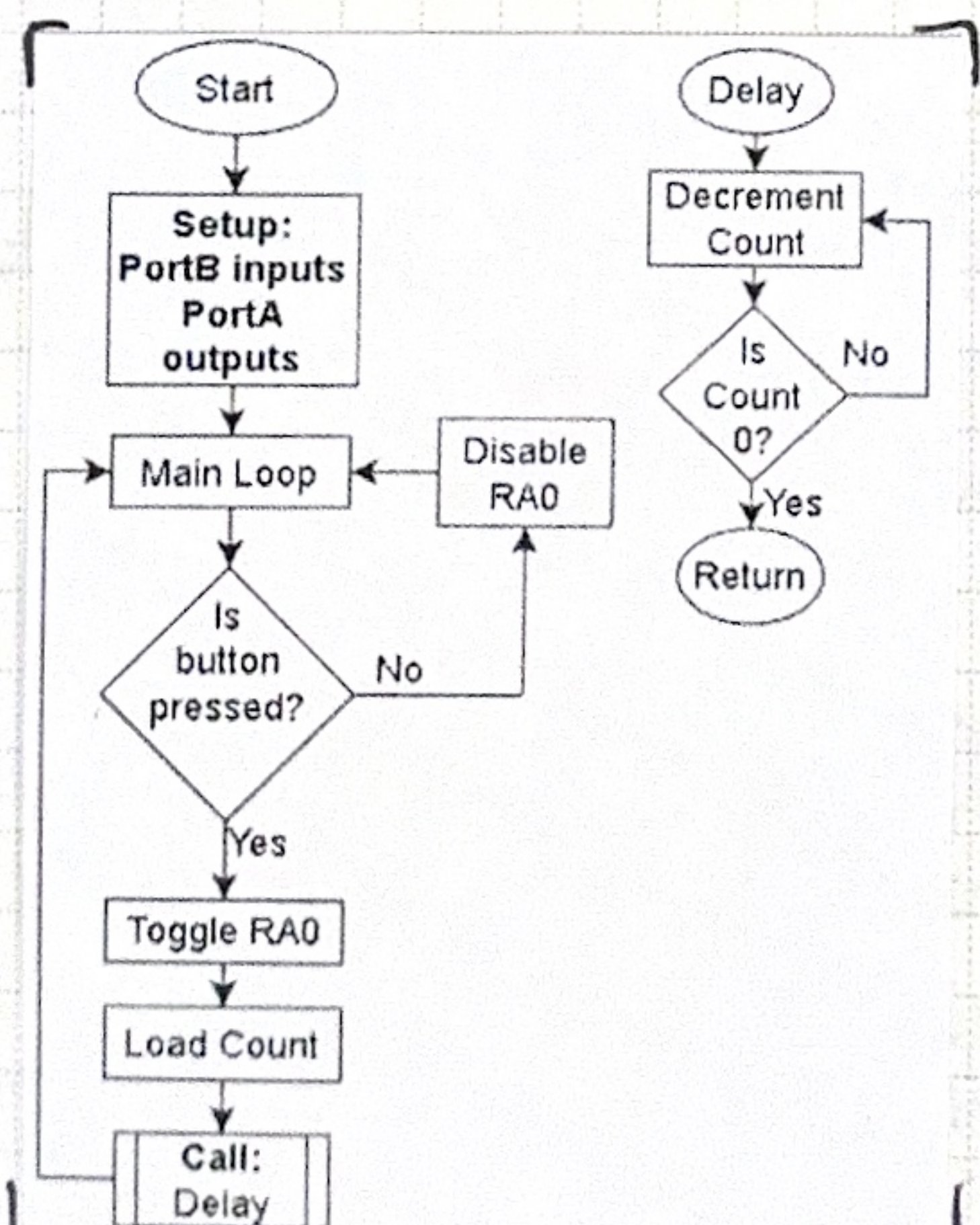


Figure 1-2 O-scope Capture of RAO output

Agilent 53131A 225 MHz Universal Counter

1,000.07007 Hz

Freq Counter of RAO



Measured Values

freq = 1.00007 kHz
 Period = 999.930 μ s
 PW = 499.9732 μ s
 PS = 499.954 μ s
 Duty cycle = 50.0006 %

freq Counter

freq = 1.00009 kHz
 Period = 999.91 μ s
 PW = 499.96 μ s
 PS = 499.95 μ s
 Duty cycle = 50 %

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FCM for code 1-1

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Part 2 - Select Notes With DIP Switches & Lookup Table

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Reuse the priority switch - scanning work from earlier labs to select musical notes from a program memory lookup table

Note Selection

Switch	Note index	Note	Frequency (Hz)	Delay value (Hex)
RB0	1	C3	130.81	EE
RB1	2	D3	146.83	D4
RB2	3	E3	164.81	BD
RB3	4	F3	174.61	B2
RB4	5	G3	196.00	9F
RB5	6	A3	220.00	8E
RB6	7	B3	246.94	7E
RB7	8	C4	261.63	77

Table 2-1 Note Selection Table

Before Lab

Reuse & adapt the priority-switch scan code to determine the active note & store that data in a general Register

Create a lookup table to retrieve the correct hex Delay value

The lookup process should just update the Delay count.

Do not change the provided hex values & don't make a different delay loop for individual notes

*DIP Switch layout same as pg 151, Only difference is the Speaker Driver circuit on pg 171

*It is important to Note, this code uses the PCL which are the lower 8-bits of the program counter. Adding to this register causes us to jump forward in the program memory to a desired instruction. See Code 2-1

Frequency (Hz)	PW
130.81	3.8223ms
146.83	3.4053ms
164.81	3.0338ms
174.61	2.8635ms
196.00	2.5510ms
220.00	2.2727ms
246.94	2.0248ms
261.63	1.9111ms

Table 2-2 Freq to PW for Table 2-1

Finding the best I value

$$U(3I+4)+27 = 1.9111\text{ms} \quad U=119 \text{ or } 77h$$

$$(119)(3I+4)+27 = 1911.1\mu s$$

$$I = 3.944 \text{ Round to } 4$$

U will be whatever is assigned by Lookup Table

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```

Main:
// main goes here
CALL TestDIP          ;2 TCY Goes to testing dip status

BTFSC tableAddress, 7 ;1 (2) 2 TCY if it is 0 it skips
goto Main             ;2 TCY

CALL LookupTable       ;2 TCY Goes to get the address from the lookupTable
MOVWF outCount         ;1 TCY puts the address into the outerloop count

CALL Delay             ;2 TCY Goes to delay

MOVLW 0x01             ;Put a 1 in W
XORWF PORTA, 1         ;Toggle PortA

goto Main

TestDIP:
Test7:
BTFSS 0x06, 7          ;Skip next instruction if RB7 is high
goto Test6             ;If RB7 low move to next bit
goto SetAddress7; If RB7 high display 7

SetAddress7:
MOVLW 0x01             ;Select the character set for display 7
MOVWF tableAddress     ;Send W to PortC
RETURN

LookupTable:
MOVWF tableAddress, 0  ;1 TCY put known address into W
ADDWF PCL, 1           ;2 TCY Add known W with the Program Counter

NOP
RETLW 0x77             ;2 TCY selects 361.63Hz
RETLW 0x7E             ;2 TCY selects 246.94Hz
RETLW 0x8E             ;2 TCY selects 220.00Hz
RETLW 0x9F             ;2 TCY selects 196.00Hz
RETLW 0xB2             ;2 TCY selects 174.61Hz
RETLW 0xBD             ;2 TCY selects 164.81Hz
RETLW 0xD4             ;2 TCY selects 146.83Hz
RETLW 0xEE             ;2 TCY selects 130.81Hz

Delay:
OutDelay:
MOVLW 0x0A, 4          ;1 TCY
MOVWF inCount          ;1 TCY

InDelay:
DECFSZ inCount         ;1(2) TCY
goto InDelay           ;2 TCY

DECFSZ outCount        ;1(2) TCY
goto OutDelay          ;2 TCY

RETURN                 ;2 TCY
    
```

* Follow the same Delay timing equation method from pg 166

$$3(I-1) + 2 + 2$$

non 0 loop overhead

$$(U-1)(3I+4) + (3I+3) + 2 + 1$$

non 0 loop Return overhead

$$U(3I+4) - 1 + 2 + 2 + 1 + 5 + 2$$

loop Return call Lookup call overhead

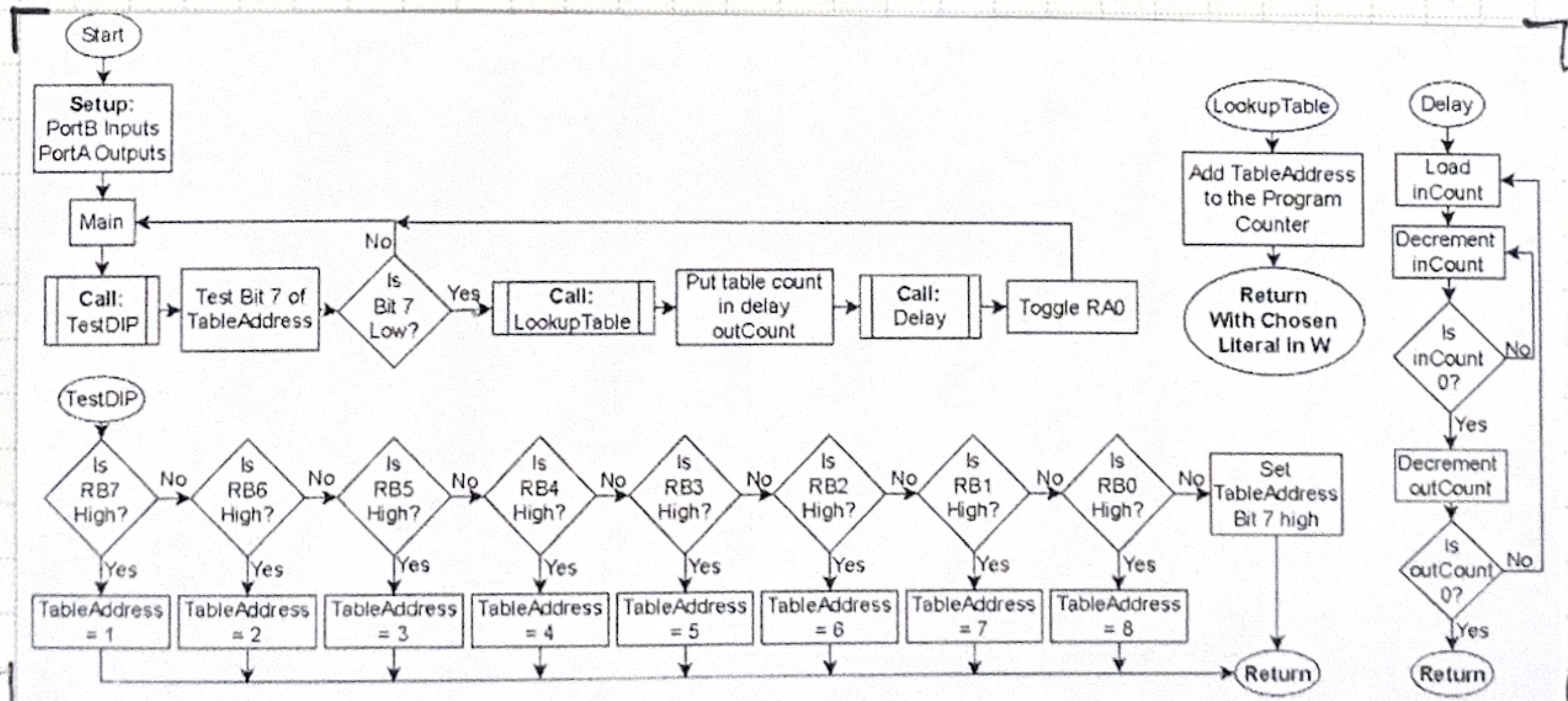
$$U(3I+4) + 11 + 2 + 2 + 2 + 4 + 2 + 2 + 2$$

BTFSC Toggle PortA Goto main Call Test DIP & Return BTFSS Goto main set monWF

$$U(3I+4) + 27 \rightarrow \text{use on pg 174}$$

* This is highly dependant on which switch is on. lower priority testing adds Tcy. It is however, insignificant to the frequency the speaker outputs as human ears cant hear a 30µs difference

Code 2-1 Lookup table & Delay



FC 2-1 for Code 2-1

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In the Lab

1. Verify the no-switch condition first so the tone output remains low
2. verify each switch individually, & they play the assigned Tone
3. Test multiple switches to confirm priority
4. Measure a low note & a high note on the O-scope & freq counter
5. Compare measured frequencies with Table 2-1
6. Document Trouble shooting

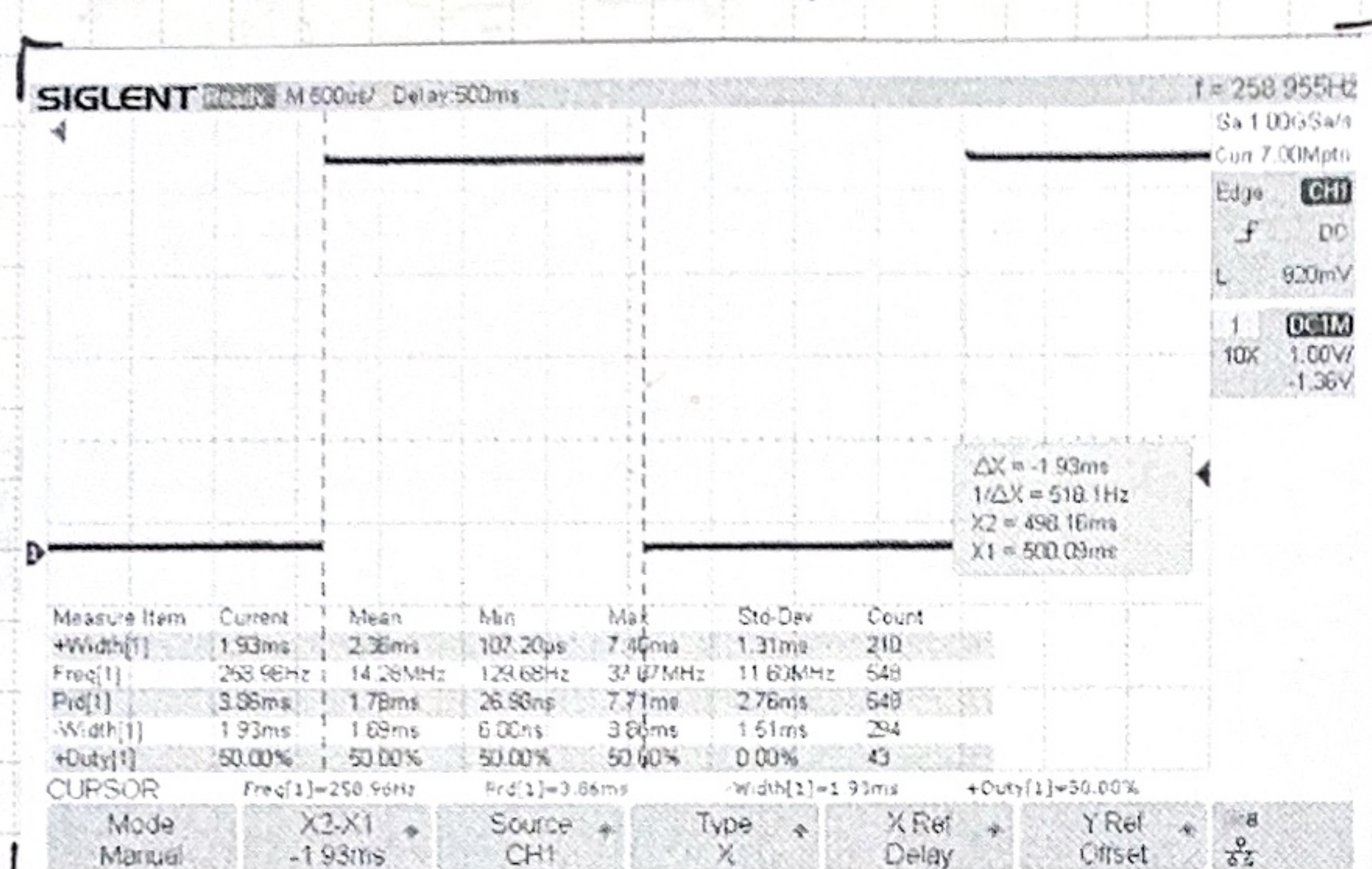


Figure 2-1 O-Scope of RB7 active Note High

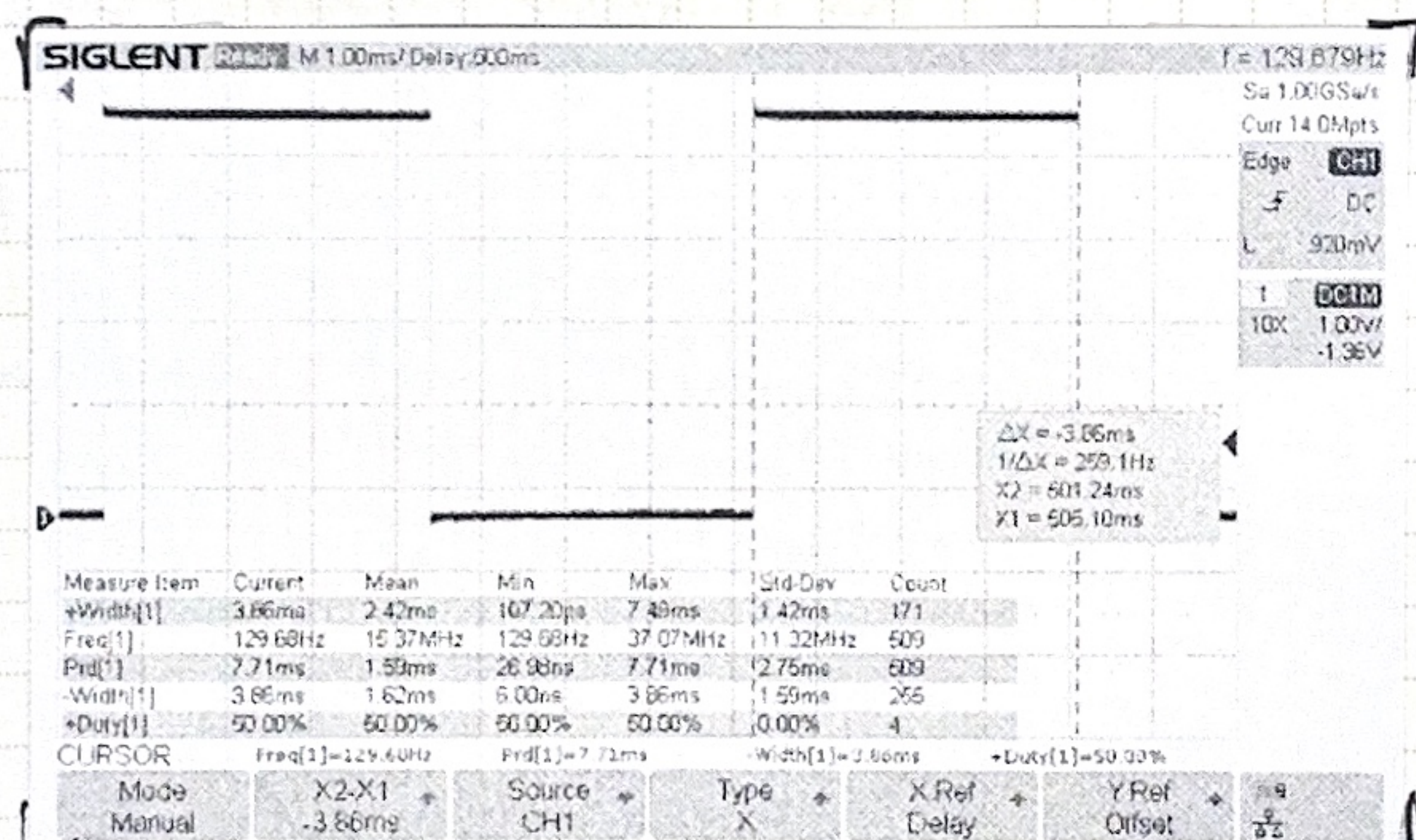


Figure 2-3 O-Scope of RB0 Active Note Low

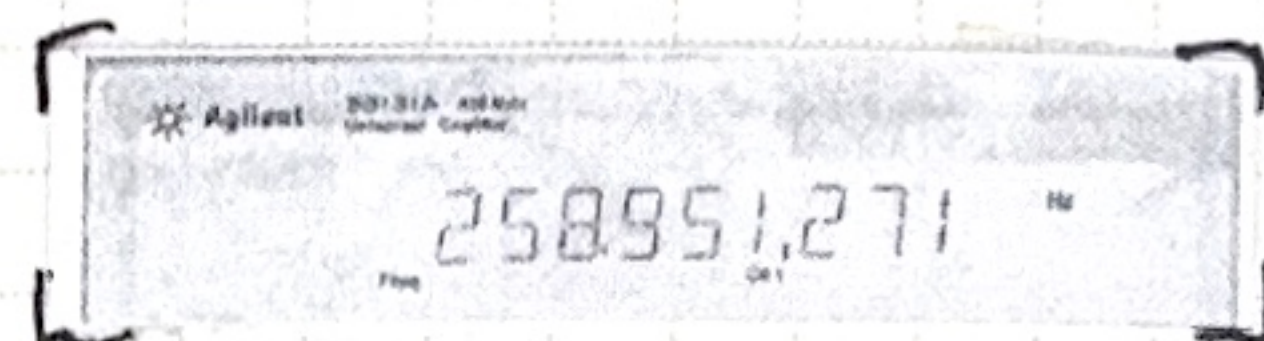


Figure 2-2 Freq Counter of RB7 Active note

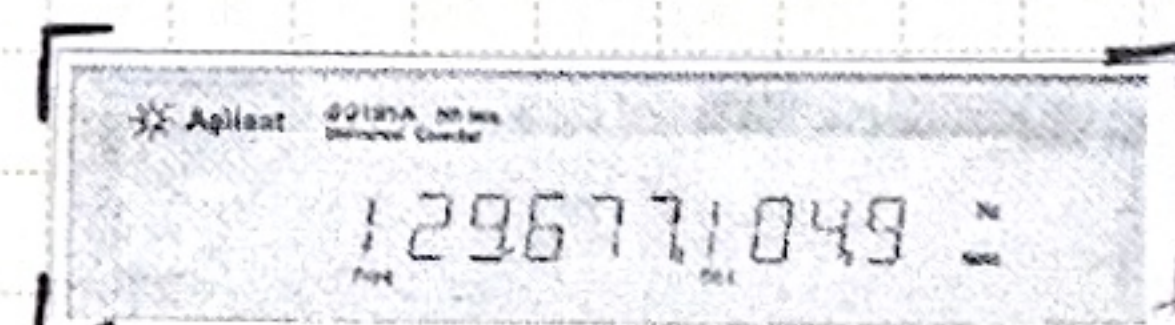


Figure 2-4 Freq Counter of RB0 Active Note

Quantity	Measured Low note Oscilloscope	Measured High note Oscilloscope	Measured Low note Frequency Counter	Measured High note Frequency Counter
PW	3.86ms	1.93ms	3.8557ms	1.9309ms
PS	3.86ms	1.93ms	3.8557ms	1.9309ms
Period	7.71ms	3.86ms	7.71146ms	3.8617ms
Frequency	129.68Hz	268.258.96Hz	129.6771Hz	258.9513Hz
Duty cycle	50%	50%	50.0000%	50.0001%

Part 3 - Build the 16-Key Muzak KeyboardGoal

Replace the DIP-switch inputs with the 4x4 Keypad Scanner & Code

Use This Key Mapping

Physical key	Note index	Note	Frequency (Hz)	Delay value (Hex)
1	1	C3	130.81	EE
2	2	D3	146.83	D4
3	3	E3	164.81	BD
A	4	F3	174.61	B2
4	5	G3	196.00	9F
5	6	A3	220.00	8E
6	7	B3	246.94	7E
B	8	C4	261.63	77
7	9	D4	293.66	6A
8	10	E4	329.63	5E
9	11	F4	349.23	59
C	12	G4	392.00	4F
*	13	A4	440.00	47
0	14	B4	493.88	3F
#	15	C5	523.25	3B
D	16	D5	587.33	35

Priority is now based on this list, not the previous lab order

Table 3-1 Note Selection map & Target frequencies

Before Lab

Reuse the Keypad circuit setup (pg 156) & scanning algorithm. The only difference is adding the driver circuit (pg 171).

Tone generation shouldn't change & no note specific adjustments are necessary.

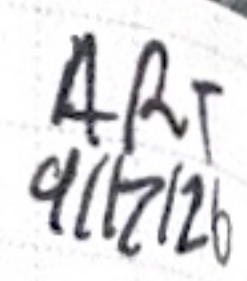
* Just like the last Keypad code, only one row pin at a time will be made an output. The only difference is the priority order of the scan. (pg 155)

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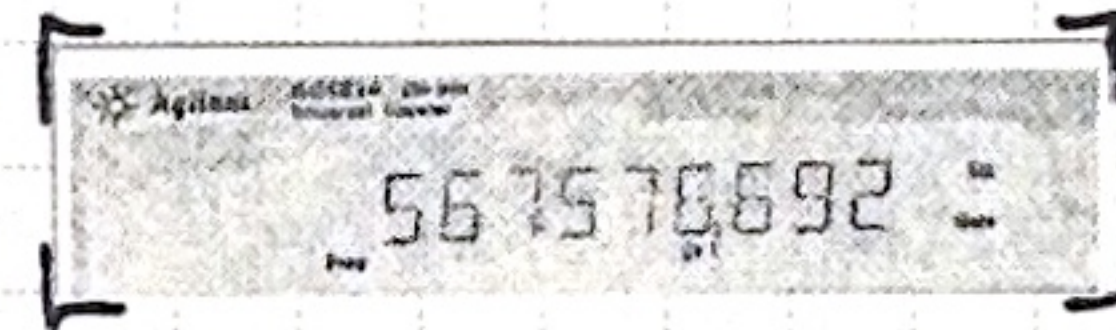
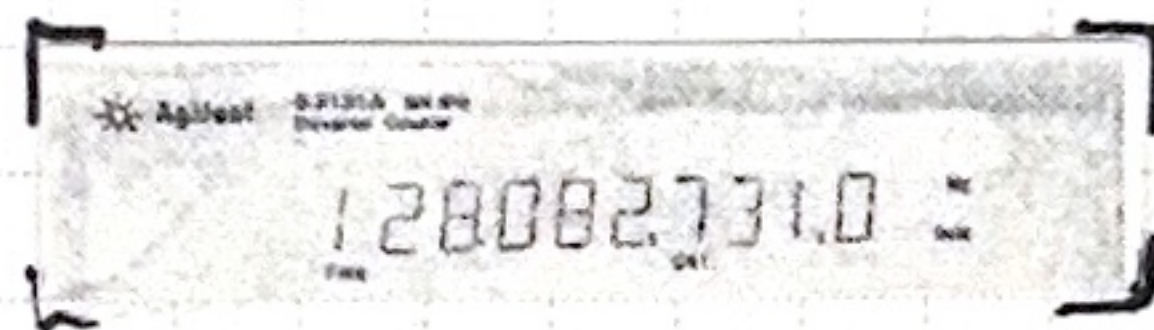
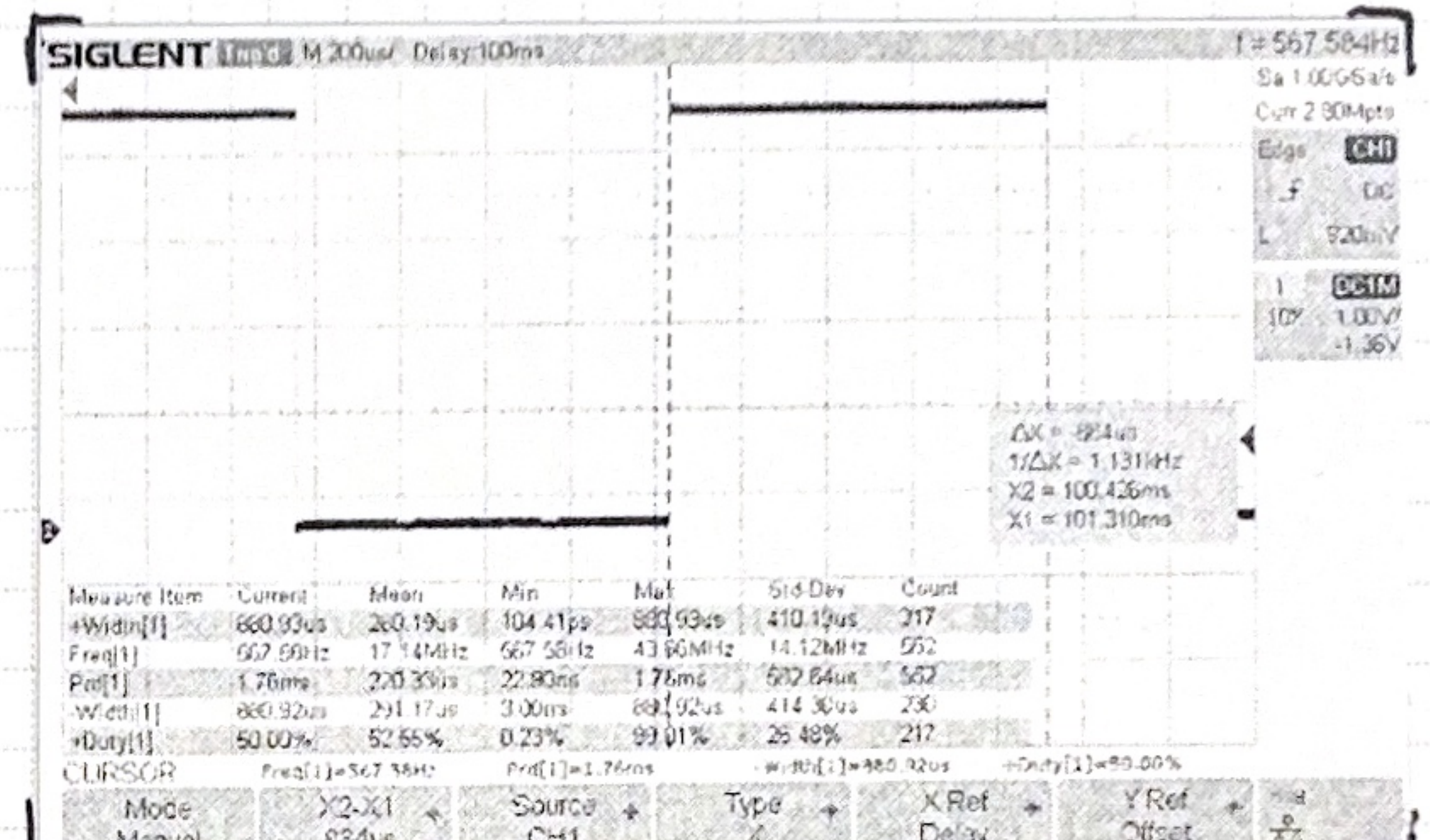
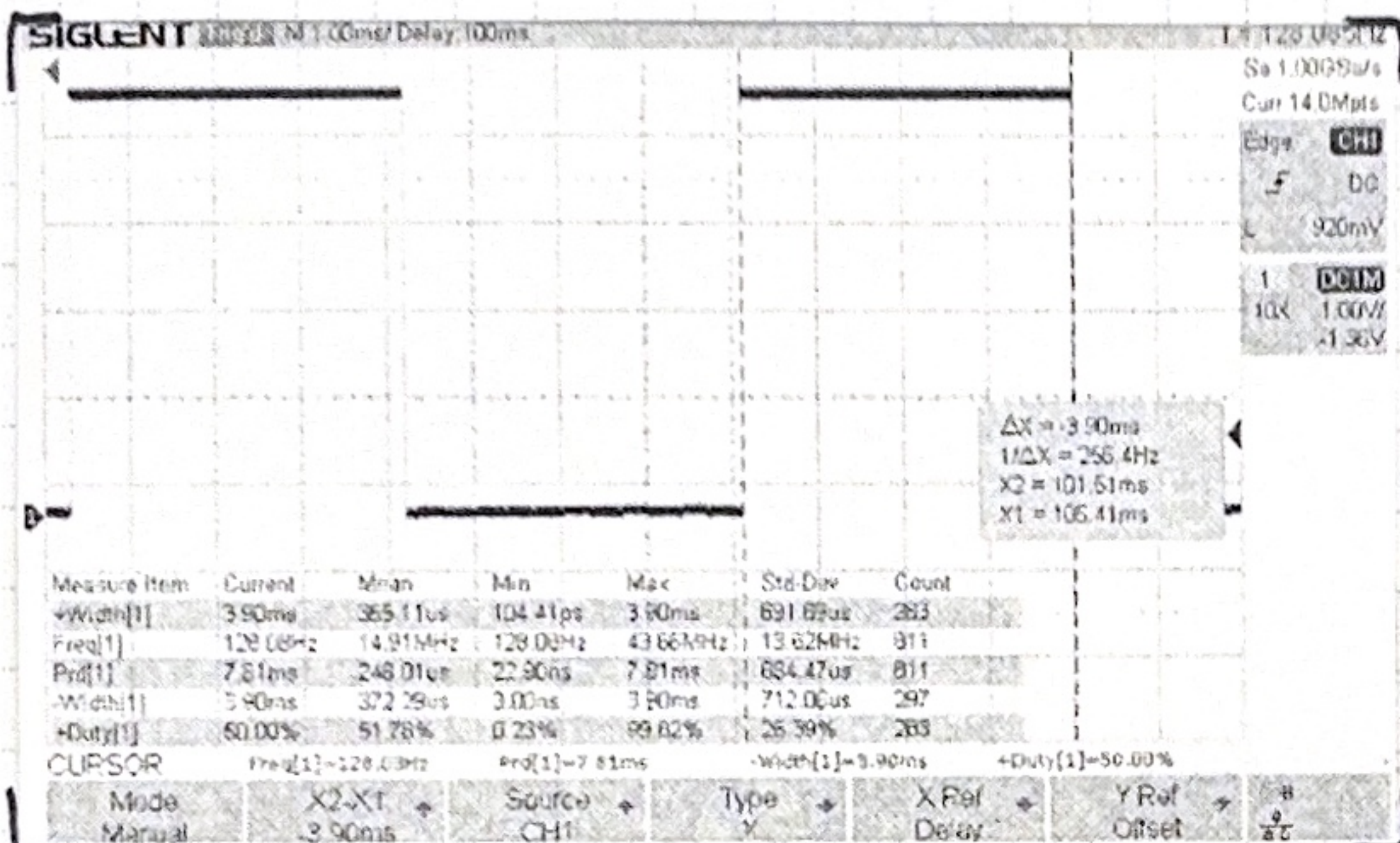


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In the Lab

1. Connect the keypad using previously verified interface (pg 156 for keypad) (pg 171 for speaker)
2. Confirm Nonoise function & key priority.
3. Measure highest & lowest notes on O-Scope & freq Counter.



Quantity	Measured Low note Oscilloscope	Measured High note Oscilloscope	Measured Low note Frequency Counter	Measured High note Frequency Counter
PW	3.9ms	880.93μs	3.9037ms	880.944μs
PS	3.9ms	880.92μs	3.9037ms	880.932μs
Period	7.81ms	1.76ms	7.807 Hz	1.76187ms
Frequency	128.08Hz	567.58Hz	128.0827 Hz	567.577 Hz
Duty cycle	50%	50%	50.00008%	50.0003%

Table 3-2 Measured high & Low Notes

Conclusion

This Lab had only one new concept in coding. That is the Lookup table which used the program counter PCH Register to jump forward to a specific instruction. Other than that most of these concepts were reused from previous Labs. Git URL's: Lab4(pg 171) Lab2(447) Lab3(159)

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